1. Problem Statement

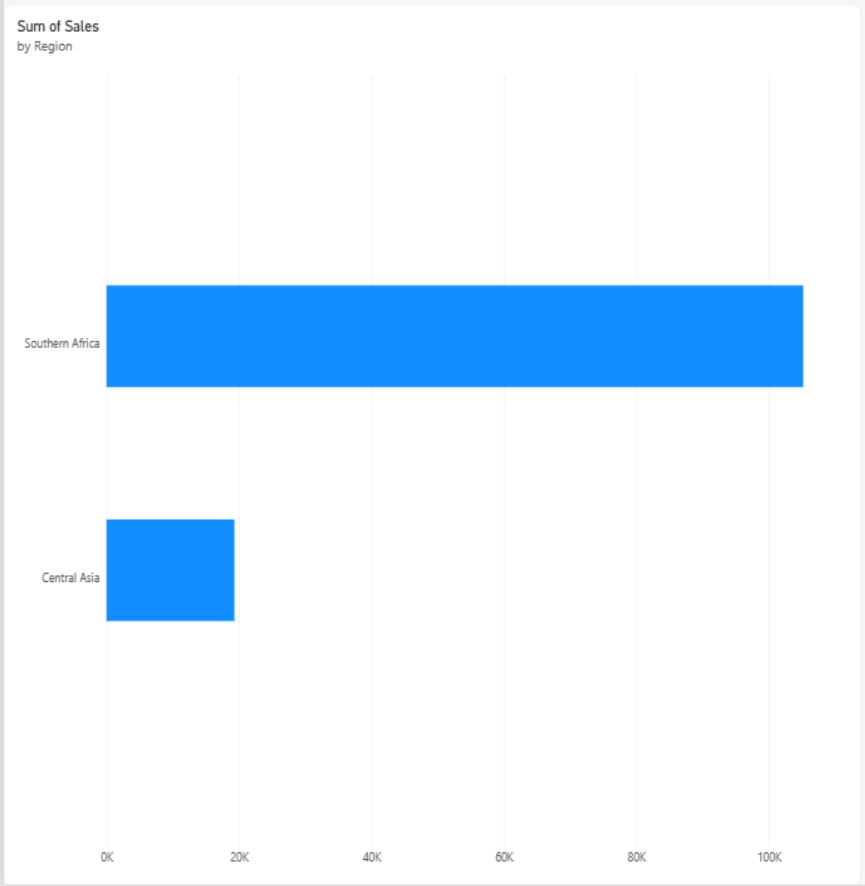
Business Intelligence systems contain sensitive organizational information related to sales, finance, customers and regional performance. Without proper authentication, authorization and access control mechanisms, unauthorized users may access confidential data, misuse information or view records beyond their responsibilities. Therefore, there is a need to design a secure BI access management system using Power BI by defining user roles, authentication methods, permission matrices and Row-Level Security (RLS), along with identifying security risks and implementing appropriate security controls to protect business information.

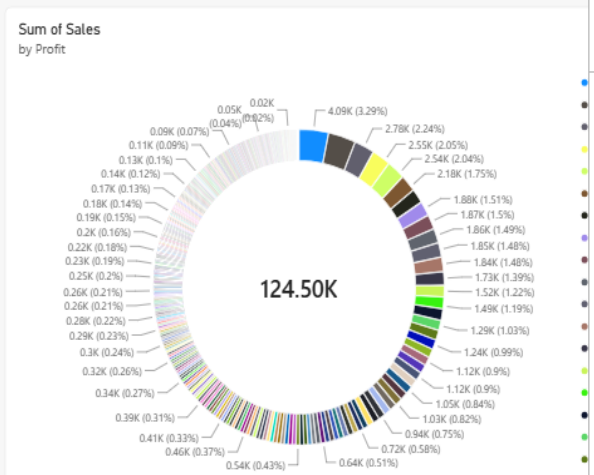
2. Solution

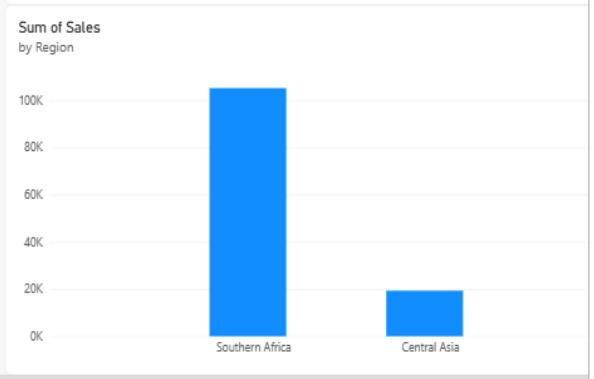
- Created a Role_Matrix to define organizational roles and their report access requirements.
- Developed an Authentication_Analysis table to compare username-password, organizational account and multi-factor authentication methods.
- Created an Access_Control_Matrix to define View, Edit, Share and Admin permissions for different users.
- Implemented a region-based security role named Southern_Africa_Manager using Row-Level Security (RLS).
- Applied a filter on the Orders table to restrict access to Southern Africa records.
- Tested the RLS role using the Modeling → View as option.
- Verified that only Southern Africa sales data was displayed when the security role was active.
- Created a Security_Risk_Assessment table to identify risks such as unauthorized access, data leakage and weak authentication.
- Developed a Security_Recommendations table containing suitable security controls and preventive measures.
- Tested and verified the security configuration using Power BI Desktop.

Now viewing as: Central Asia Manager, South Africa Manager

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